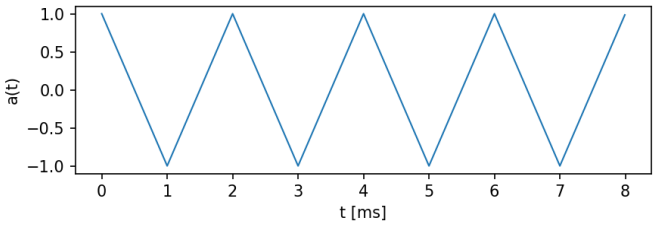
EXERCISE 1

FFT of a Triangle Signal

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Today's exercise gives you insight into the FFT and its applications. We want to compute the DFT of the following triangle signal and study various properties of the FFT along the way.



1 DFT.

a) Implement a naive DFT and inverse DFT and visualize the results $\tilde{a}(\omega)$ and the decomposition into individual Fourier components.

$$\hat{a}_k = \sum_{j=0}^{N-1} a_j \, e^{-2\pi i \, jk/N}, \qquad a_j = \frac{1}{N} \sum_{k=0}^{N-1} \hat{a}_k \, e^{+2\pi i \, jk/N}$$

b) Compare with numpy's FFT.
Why is the difference not exactly zero?

c) Compare timings of DFT and FFT for varying system size N.

d) Compare with the analytic result.

- How do Δt and T influence the FFT result?
- Why does the error grow towards the Nyquist frequency?

2 Properties of the DFT (verify numerically).

a) Verify the symmetry $\hat{a}_{N-k} = \hat{a}_k^*$ for real signals.

b) Shift the signal and compare with the shift theorem $\hat{a}_k \, e^{-2\pi i \, km/N}$.
Why is $\hat{a}_k$ even purely real for the triangle?

c) Check $\mathcal{F} \circ \mathcal{F}^{-1}(a) = a$ and Parseval's theorem $\sum_j |a_j|^2 = \sum_k |\hat{a}_k|^2$.

d) Check the recursion (Cooley-Tukey) numerically:

$$\begin{aligned} \hat{a}_k &= E_k + w_N^{-k} O_k \\ \hat{a}_{k+N/2} &= E_k - w_N^{-k} O_k \end{aligned} \qquad E_k = \mathrm{DFT}(a_0, a_2, \dots), \quad O_k = \mathrm{DFT}(a_1, a_3, \dots), \quad w_N = e^{2\pi i/N}$$

e) Based on this, implement a recursive FFT (radix-2).

3 Try out different signals (rectangle, sawtooth, sine, Gaussian pulse, noise, ...).

a) Which spectra are real, imaginary, complex?

b) How is the decay of $|\hat{a}_k|$ related to smoothness (jumps, kinks)?

c) What happens if the signal does not fit into the sample window? How can this be improved?